

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 5-1

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1. TITLE

Interactive Product Calculator

2. AIM

To create an interactive Product Calculator webpage that accepts a product name, quantity, and price, and dynamically calculates and displays the total price using JavaScript functions and event listeners without refreshing the webpage.

3. OBJECTIVE

- To understand JavaScript functions for performing calculations.
- To implement DOM manipulation for reading input values and updating the webpage.
- To use `addEventListener()` to respond to changes in quantity and price.
- To calculate total price dynamically using the formula $\text{Quantity} \times \text{Price}$.
- To display the updated result without refreshing the webpage.

4. THEORY

- JavaScript functions allow reusable blocks of code to perform specific tasks such as calculations.
- DOM Manipulation enables JavaScript to access and modify HTML elements dynamically.
- Event Handling allows the webpage to respond to user actions such as changing input values.
- The input event is used to detect changes in the quantity and price fields.
- Methods such as `getElementById()`, `textContent`, and `addEventListener()` are used to create an interactive webpage.

Flow of application: User Input → DOM Selection → JavaScript Function → Calculation → Dynamic Total Display

5. ALGORITHM / PROCEDURE

1. Create an HTML file with input fields for Product Name, Quantity, and Price.
2. Add a section to display the calculated Total Price.
3. Use CSS to style the calculator interface.
4. Select the input and output elements using document.getElementById().
5. Define a calculateTotal() JavaScript function.
6. Read the quantity and price values from the input fields.
7. Multiply quantity by price to calculate the total price.
8. Use addEventListener() with the input event for the quantity and price fields.
9. Display the calculated total dynamically using textContent.
10. Test the calculator with different product names, quantities, and prices.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>QuickCart - Product Calculator</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <main class="page">

    <section class="card">

      <h1>QuickCart Product Calculator</h1>

      <label for="productName">Product Name</label>
      <input
        type="text"
        id="productName"
        placeholder="Enter product name"
```

```

    >

    <label for="price">Price</label>
    <input
      type="number"
      id="price"
      placeholder="Enter price"
      min="0"
    >

    <label for="quantity">Quantity</label>
    <input
      type="number"
      id="quantity"
      placeholder="Enter quantity"
      min="1"
      value="1"
    >

    <div class="result">
      <h2>Total Price</h2>
      <p>₹<span id="total">0.00</span></p>
    </div>

  </section>

</main>

<script src="script.js"></script>

</body>
</html>

```

script.js

```

const productName = document.getElementById("productName");
const price = document.getElementById("price");
const quantity = document.getElementById("quantity");
const total = document.getElementById("total");

function calculateTotal() {
  const productPrice = parseFloat(price.value) || 0;
  const productQuantity = parseInt(quantity.value) || 0;

  const totalPrice = productPrice * productQuantity;

  total.textContent = totalPrice.toFixed(2);
}

```

```
price.addEventListener("input", calculateTotal);
quantity.addEventListener("input", calculateTotal);
productName.addEventListener("input", calculateTotal);
```

Style.css

```
* {
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background-color: #f2f2f2;
  margin: 0;
  padding: 40px;
}

.page {
  display: flex;
  justify-content: center;
}

.card {
  width: 400px;
  background-color: white;
  padding: 25px;
  border-radius: 10px;
}

h1 {
  text-align: center;
}

label {
  display: block;
  margin-top: 15px;
  font-weight: bold;
}

input {
  width: 100%;
  padding: 10px;
  margin-top: 5px;
}

.result {
  margin-top: 25px;
  padding: 15px;
  text-align: center;
```

```

    background-color: #eeeeee;
}

.result p {
    font-size: 25px;
    font-weight: bold;
}

```

7. CODE EXPLANATION

Code	Explanation
document.getElementById()	Accesses the product name, quantity, price, and total elements.
calculateTotal()	Defines the function that calculates the total price.
Number()	Converts quantity and price input values into numbers.
qty * unitPrice	Calculates the total price using Quantity × Price.
addEventListener("input")	Runs the calculation whenever quantity or price changes.
textContent	Displays the updated total directly on the webpage.

8. TEST CASES AND EXECUTION DATA

Test Case	Input	Expected Result	Actual Result	Status
TC-01	Product: Pen; Qty: 2; Price: ₹10	Total ₹20.00	Works correctly	Pass
TC-02	Product: Book; Qty: 3; Price: ₹50	Total ₹150.00	Works correctly	Pass
TC-03	Change quantity from 2 to 5	Total updates immediately	Works correctly	Pass
TC-04	Change price from ₹10 to ₹15	Total updates immediately	Works correctly	Pass
TC-05	Qty: 4; Price: ₹25.50	Total ₹102.00	Works correctly	Pass

9. OUTPUT

The webpage displays the Product Name, Quantity, Price, and dynamically calculated Total Price.

QuickCart Product Calculator

Product Name

Enter product name

Price

Enter price

Quantity

1

Total Price

₹0.00

QuickCart Product Calculator

Product Name

purple

Price

822

Quantity

100

Total Price

₹82200.00

10.Result

Thus, an Interactive Product Calculator was successfully developed using HTML, CSS, and JavaScript. The application dynamically calculates and displays the total price whenever the quantity or price changes, without refreshing the webpage. The experiment demonstrates JavaScript functions, DOM manipulation, event handling, and dynamic webpage updates.